

Introduction

I. Visual Impairment and Blindness

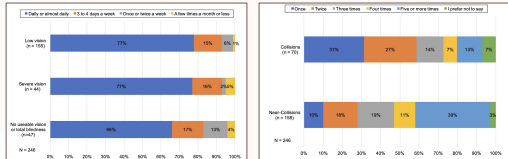
A. Legal blindness: visual acuity < 6/60

- ~1,000,000 legally blind individuals in US (NIH) ("Vision," 2022)

II. Challenges of Blind Navigation

A. Walking navigation → hazardous (Hassan, 2012)

- Unreliable crossing decisions (Hassan, 2012)



III. Existing Solutions

A. Human guides (Garaj et al., 2003)

- Restrict independence, unreliable availability

B. Guide dogs - \$\$\$ (Jeamwathanachai et al., 2019)

C. White canes (Jeamwathanachai et al., 2019)

- Elevated obstacles

IV. Drawbacks of Existing Solutions

A. Treatments for blindness

- only considered experimentally

(Garaj et al., 2003)

B. biped.ai: \$1990



Fig. 2. biped.ai Apparatus

V. Neural Networks

A. Object detection capabilities

B. YOLOv4-tiny

- Efficient

→ portable,

ideal for edge

computing

(Asad et al., 2022)

- Highest FPS

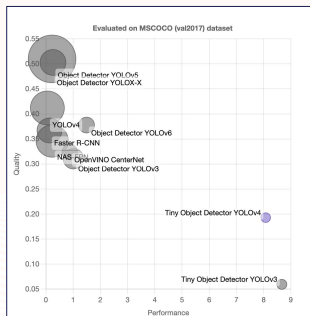
compared to

other models

(Asad et al., 2022)

- Open-source

model



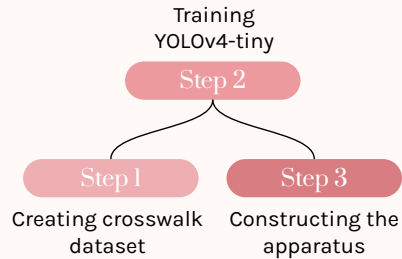
Objectives

- Develop a YOLOv4-tiny model trained on a **custom dataset** to **assist pedestrians** in real-time during **crosswalk navigation**
- Construct a **practical, user-friendly wearable apparatus** that implements the aforementioned model on a Raspberry Pi

Construction of an Affordable Wearable Apparatus for Visually Impaired Pedestrians

Brian Liu, Michelle Luo, Maggie Wu

Methods Overview



YOLOv4-tiny Model

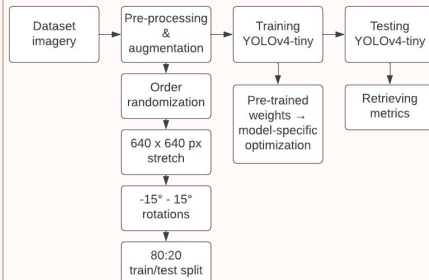


Fig. 5. Workflow of constructing YOLOv4-tiny model

YOLOv4-tiny Results

Confusion Matrix

	Actual	
	true	false
Predicted	TP: 780	FP: 54
	FN: 34	TN: N/A

Metrics

$$mAP = \frac{1}{k} \sum_i AP_i$$

$$precision = \frac{TP}{TP + FN}$$

$$recall = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2 * precision * recall}{precision + recall}$$

Training Results

- mAP: **0.92**
- Precision: **0.94**
- Recall: **0.96**
- F₁-score: **0.95**



Fig. 6. Sample image showing YOLOv4-tiny's annotations

Custom Dataset

I. Creation

A. Recording videos

- Taken in Great Neck
- Crosswalk + signal

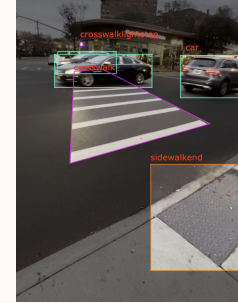
B. Data annotation on Roboflow (Azmi et al., 2022)

II. Splitting the dataset

A. 2800+ frames extracted

- Addition of other datasets → **6000+** total frames

Fig. 4. Annotated frame extracted from a video.



Wearable Apparatus



Fig. 7. Labeled diagrams of the apparatus

User Survey Results

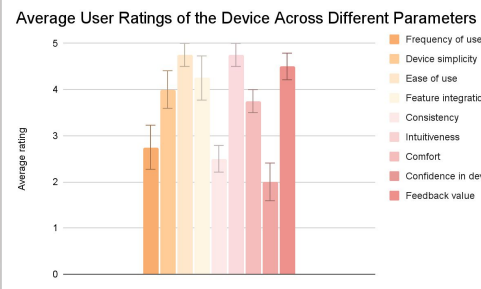


Fig. 8. Average user ratings of the apparatus across various parameters, such as simplicity, consistency, and comfort (n=4). Users were asked to evaluate each variable on a scale from 1 to 5. Questions incorporated the SUS questionnaire (Brook 1996). The value of each bar represents +/- SE.

Discussion

- The first goal was to develop a fast and accurate YOLOv4-tiny model trained on a custom dataset
 - This model achieved a mean average precision (mAP) of 0.92
 - Comparable to results of studies with other YOLOv4-tiny models, which achieved mAPs of 0.78, 0.80, and 0.94 (Jiang et al., 2022; Asad et al., 2022; Guo et al., 2021)
- The second goal was to construct a wearable device that implements the model in a user-friendly way
 - Device simplicity, ease of use, and feedback value averaged at least 4 out of 5
 - Consistency received one of the lowest scores, with an average rating of 2.5 out of 5
 - This explains the lower ratings of confidence in the device and willingness to use it frequently
 - The apparatus may thus work best as a supplement to existing forms of navigation assistance such as white canes

Limitations and Future Research

• Limitations

- Functional in limited lighting conditions
- Did not survey fully blind testers who regularly use white canes

• Future Research

- More variation of lighting conditions in dataset
- Compare different object detection models
- Decrease delay between observations and feedback

Conclusion

- This study accomplished the original goals to:
 - Develop a YOLOv4-tiny model trained on a custom dataset to assist pedestrians in real-time during crosswalk navigation
 - Construct a practical, user-friendly wearable apparatus that implements the aforementioned model on a Raspberry Pi
- The model had a mAP of 0.92, comparable to other YOLOv4-tiny models deployed in edge computing
- Users agreed that the device was easy to use and provided valuable feedback on their surroundings

References

- Asad, M. H., Khalid, S., Yousef, M. H., Ullah, M. G., & Ahmad, A. (2022, April 20). Pedestrian detection using Deep Learning: A real-time and ai-on-the-edge perspective. *Advances in Civil Engineering*. Retrieved April 16, 2023, from <https://www.civildatas.com/journal/article/2022/04/20/20220420>
- Azmi, N. N., Sophan, A., & Sawana, A. A. (2022). Deep-learning-based detection of missing road lane markings using YOLOv5 algorithm. *IOP Conference Series: Materials Science and Engineering*, 1244(1), 012021. <https://doi.org/10.1088/1757-8999/1244/1/012021>
- Brooke, J. (1996). SUS-A quick and dirty usability scale. *Usability evaluation in industry*, 18(194), 4-7.
- Garaj, V., Jirawattana, R., Piatkowski, P., Cocchi, F., & Balachandran, W. (2003). A system for remote sighted guidance of visually impaired pedestrians. *British Journal of Visual Impairment*, 21(2), 55-61. <https://doi.org/10.1177/0264619603026024>
- Jiang, C., Ren, H., Ye, X., Zhu, J., Zeng, H., Nan, Y., Sun, M., Ren, X., & Huo, H. (2022, July 18). Object detection from UAV thermal infrared images and videos using YOLO models. *International Journal of Applied Earth Observation and Geoinformation*, 112, 102822. <https://doi.org/10.1016/j.isprsjis.2022.102822>
- Hassan, S. E. (2012, May 4). Are Normally Sighted Visually Impaired and Blind Pedestrians Accurate and Reliable at Making Street Crossing Decisions? *Investigative Ophthalmology & Visual Science*, 53(6), 2593. <https://doi.org/10.1167/jov.11.04.0460>
- Jeamwathanachai, W., Wad, M., & Willis, G. (2019). Indoor navigation by blind people: Behaviors and challenges in unfamiliar spaces and buildings. *British Journal of Visual Impairment*, 37(2), 140-153. <https://doi.org/10.1177/0264619618833223>
- Jiang, C. C., Ren, H., Ye, X., Zhu, J., Zeng, H., Nan, Y., Sun, M., Ren, X., & Huo, H. (2022). Object detection from UAV thermal infrared images and videos using YOLO models. *International Journal of Applied Earth Observation and Geoinformation*, 112, 102822. <https://doi.org/10.1016/j.isprsjis.2022.102822>
- Liu, S., Fitzharris, M., & Orling, J. (2018). The impact of electric/hybrid vehicles and bicycles on pedestrians who are blind or have low vision. *Monash University*. Retrieved December 4, 2022, from https://research.monash.edu/en/portals/132000360/30300360_no_oad/
- Tiny Object Detector YOLOv4. *modelscope.ai*. (n.d.). Retrieved January 19, 2023, from <https://modelscope.ai/models/32>
- Visual impairment and blindness. (2022, October 13). <https://www.bbc.com/news/health-62464111>
- Visual impairment, blindness cases in U.S. expected to double by 2050. (2018, May 19). National Institutes of Health (NIH). <https://www.nih.gov/news-events/news-releases/visual-impairment-blindness-cases-us-expected-double-2050>